SOMASHEKHAR JAVOORU

Bengaluru, KA | somashekharjavooru@gmail.com | +918971941346 | LinkedIn | GitHub

Summary

Embedded Software Developer and 2026 ECE graduate with hands-on experience in firmware, C/C++, and Python on Linux and RTOS (FreeRTOS) platforms. Experienced in peripheral drivers and interfacing over I2C, SPI, and UART, system bring-up, interrupt-driven programming, and debugging performance and reliability issues. Solid grounding in Linux kernel fundamentals, networking (Ethernet, TCP/IP, MQTT, L2/L3 concepts), and Operating Systems, with a strong interest in networking platforms.

Education

Bapuji Institute of Engineering and Technology (Davanagere) *Aug 2022 – Jun 2026*
Bachelor of Engineering in Electronics and Communication | **CGPA: 8.75/10.0**
Coursework: Operating Systems, Linux Internals, Linux Kernel Fundamentals, Networking Concepts, Data Structures & Algorithms, Object-Oriented Programming

SPANDANA PU SCIENCE COLL, KULKARNIPET, ILKAL *Pre-University Course (PUC)*
Science | Karnataka Department of Pre-University Education | **86.6%**

Technical Skills

- **Firmware & Systems:** Embedded Software Development, Embedded Linux, Firmware Development, RTOS (FreeRTOS), Peripheral Drivers (I2C, SPI, UART), ISRs, Hardware-Software Integration, System Bring-up
- **Programming Languages:** C, C++, Python, Java
- **Networking:** Ethernet, TCP/IP, HTTP, MQTT, Wi-Fi, L2/L3 Concepts
- **Operating Systems & Fundamentals:** Linux, Linux Kernel Fundamentals, Linux System Programming, Concurrency, Memory Management, Data Structures & Algorithms, OOP
- **Engineering Practices:** Git, Unit Testing, Debugging, Root-Cause Analysis, Code Reviews, Software Integration, Agile SDLC

Experience

Dextris Infoservices Pvt Ltd *Jun 2026 – Present*
Embedded Engineer Intern
- Optimized concurrent task execution and thread scheduling using FreeRTOS, reducing CPU idle time by 20% to improve overall resource efficiency.
- Led root-cause analysis and hardware-software debugging during system bring-up and integration, reducing application boot-up latency by 1.5 seconds.
- Engineered modular C/C++ applications using OOP principles, refining execution algorithms to decrease software response time by 25% for high-speed data telemetry.
- Built an HTTP-based video streaming subsystem using Python and C++, achieving stable 30 FPS transmission across wireless networks for real-time monitoring.
- Delivered software releases using Git version control and coordinated remote Over-The-Air (OTA) firmware updates.

EchoBrains *Jan 2026 – May 2026*
Embedded Systems Intern
- Developed C/C++ control algorithms for microcontroller targets, achieving <1 ms execution latency for time-critical applications.
- Conducted unit testing and verification on continuous data streams, ensuring 0% packet loss across concurrent communication buses (I2C, SPI, UART).

Projects

Real-Time Face Recognition System | *C/C++, Python, HTTP, Wi-Fi* *Jun 2026* | Link
- Architected C/C++ firmware to interface with camera hardware, executing computer vision algorithms with <200 ms processing latency.
- Synchronized concurrent networking routines for SPI display handling and HTTP data transmission via Wi-Fi to a centralized server.
- Resolved critical memory leaks during continuous streaming by applying memory management and root-cause debugging techniques.

Automated Gas Leakage Detection System | *C/C++, MQTT, Networking, ISRs* *Mar 2025* | Link
- Deployed MQTT telemetry streaming over Wi-Fi to transmit real-time sensor data for remote monitoring, achieving 99.9% system uptime.
- Programmed event-driven interrupt service routines (ISRs) to trigger safety protocols upon hazard detection in <500 ms.

AI-Powered Agriculture Robot | *Python, C/C++, Linux, Machine Learning* *Dec 2025* | Link
- Implemented Python-based CNN image processing pipelines within a Linux environment to autonomously classify conditions, improving algorithmic routing efficiency by 15%.
- Designed C-based algorithms utilizing precise timing logic for motor control, improving automated inspection accuracy by 20%.

Certifications & Awards

- **Smart India Hackathon:** Engineered IoT and networking solutions with a 4-member cross-functional team.
- **Dextris Recognition:** Awarded for core contributions to software architecture, automated OTA deployment, and software bring-up.
- **HackerRank Java (Basic):** Certified in core programming concepts, data structures, and object-oriented logic.